Progress Check

(i) Let X be the mid-point of the line segment joining $A(3,0)$ and $B(-5,4)$ and Y be the mid-point of the line segment joining $P(-11,-8)$ and $Q(8,-2)$. Find the mid-point of the line segment XY.

(ii) If $(3,x)$ is the mid-point of the line segment joining the points $A(8,-5)$ and $B(-2,11)$, then find the value of 'x'.

Example 5.13

If $(x,3)$, $(6,y)$, $(8,2)$ and $(9,4)$ are the vertices of a parallelogram taken in order, then find the value of x and y.

Solution Let $A(x,3)$, $B(6,y)$, $C(8,2)$ and $D(9,4)$ be the vertices of the parallelogram $ABCD$. By definition, diagonals AC and BD bisect each other.

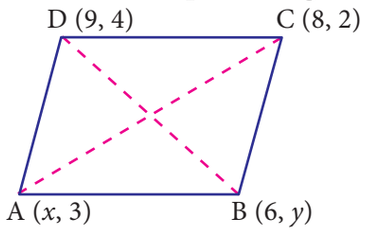


Fig. 5.25

Mid-point of AC = Mid-point of BD

$$\left(\frac{x+8}{2},\frac{3+2}{2}\right)=\left(\frac{6+9}{2},\frac{y+4}{2}\right)$$

equating the coordinates on both sides, we get

$$\frac{x+8}{2}=\frac{15}{2} \quad \text{and} \quad \frac{5}{2}=\frac{y+4}{2}$$

$$x+8=15 \qquad 5=y+4$$

$$x=7 \qquad y=1$$

Hence, $x = 7$ and $y = 1$.

Thinking Corner

$A(6,1)$, $B(8,2)$ and $C(9,4)$ are three vertices of a parallelogram $ABCD$ taken in order. Find the fourth vertex D. If (x_1,y_1), (x_2,y_2), (x_3,y_3) and (x_4,y_4) are the four vertices of the parallelogram, then using the given points, find the value of $(x_1+x_3-x_2, y_1+y_3-y_2)$ and state the reason for your result.

Example 5.14

Find the points which divide the line segment joining $A(-11,4)$ and $B(9,8)$ into four equal parts.

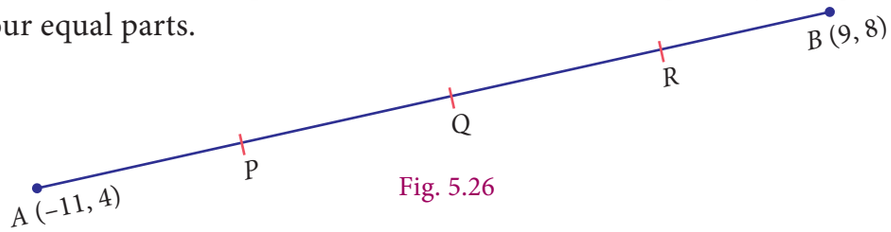


Fig. 5.26

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